

Homework Week 4-5 Modern Physics

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(Dated: March 26, 2026)

For exercises from the textbook, the corresponding number is reported. The text may have been edited for clarity.

Some of the following exercises (possibly with different numbers) may be on the **mini-test** taking place on the Monday of the following week, or on the **midterms**.

THE HYDROGEN ATOM

- **Problems from the book:** 7.6, 7.9, 7.16, 7.19

- **Hydrogen 2p orbitals**

Check by explicit calculation in Cartesian (x, y, z) coordinates that the Cartesian $2p_x$ wavefunction given in the tutorial problem set satisfies the Schrödinger equation. (This a lengthy but straightforward calculation only using the rules of differentiation.)

- **Angular momentum of 2D harmonic oscillator**

Consider the eigenstates of the 2D harmonic oscillator with potential

$$V(x, y) = \frac{m\omega^2}{2}(x^2 + y^2).$$

The eigenstates are $\psi_{nm}(x, y) = \psi_n(x)\psi_m(y)$, given as products of the 1D harmonic oscillator eigenstates ψ_n . Compute the expectation values of

$$L_z = xp_y - yp_x$$

for the ψ_{00} , ψ_{01} and ψ_{10} eigenstates, using that

$$\begin{aligned}\psi_0(x) &= \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2}, \\ \psi_1(x) &= \left(\frac{\alpha}{\pi}\right)^{1/4} \sqrt{2\alpha} x e^{-\alpha x^2},\end{aligned}$$

with $\alpha = m\omega/\hbar$. Use the integral identities introduced on tutorial 4. You can simplify many of the integrals by noting whether the integrand is even or odd.

Show that $\psi_{\pm} = \frac{1}{\sqrt{2}}(\psi_{01} \pm i\psi_{10})$ are eigenstates of L_z , i.e. $L_z\psi_{\pm} = \pm\hbar\psi_{\pm}$.

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EXTRA QUANTUM MECHANICS PROBLEMS

These are more advanced problems that are above the level we expect on the midterms, but are very useful to solve as practice. If you have any questions, feel free to contact the instructors.

• Harmonic oscillator ladder operators

It is possible to define so called “ladder” operators for the 1D harmonic oscillator, these take an energy eigenstate to another eigenstate with one $\hbar\omega$ unit higher or lower energy. The ladder operators are defined as

$$a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(x + \frac{i}{m\omega} p \right),$$

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(x - \frac{i}{m\omega} p \right),$$

where a is the ladder down (annihilation) operator, and its hermitian adjoint $a^\dagger$ is the ladder up (creation) operator.

Using explicit calculation with the differential operator form of a and $a^\dagger$, show that

$$a^\dagger \psi_0 = \psi_1,$$

$$a^\dagger \psi_1 = \sqrt{2} \psi_2,$$

$$a \psi_0 = 0,$$

where ψ_n are the eigenstates of the harmonic oscillator.

Using the position and momentum commutation relations, show the commutation relations

$$[H, a] = -\hbar\omega a,$$

$$[H, a^\dagger] = \hbar\omega a^\dagger.$$

Show that, as a consequence if ψ_n is an eigenstate of H with energy $\hbar\omega(n + 1/2)$ ($H\psi_n = \hbar\omega(n + 1/2)\psi_n$), then $a\psi_n$ is an eigenstate with eigenvalue $\hbar\omega(n - 1 + 1/2)$. Similarly show that $a^\dagger$ increases the energy by $\hbar\omega$.

Show that $H = \hbar\omega(a^\dagger a + \frac{1}{2})$. Using this, derive the general relation together with the normalization constants

$$a^\dagger \psi_n = \sqrt{n+1} \psi_{n+1},$$

$$a \psi_n = \sqrt{n} \psi_{n-1}.$$

Hint: the norm of $a\psi$ is $\langle a\psi | a\psi \rangle = \langle \psi | a^\dagger a \psi \rangle$.

• Numerical solution of the Schrödinger equation

It is possible to approximately solve the Schrödinger equation by discretizing real space, this turns the wavefunction into a finite vector, the Hamiltonian into a finite matrix, and the Schrödinger equation into a matrix eigenvalue problem.

Let $\psi_j = \psi(ja)$ for $j = 0, \dots, N-1$ the wavefunction sampled on a uniform grid with spacing a . Here we consider a one-dimensional problem, but the method generalizes to higher dimensions.

We can replace the second derivative with a finite difference approximation:

$$\frac{d^2\psi}{dx^2}(x) \approx \frac{\psi(x-a) - 2\psi(x) + \psi(x+a)}{a^2}.$$

Applying this to the position $x = ja$ we obtain

$$\left(\frac{d^2\psi}{dx^2}\right)_j \approx \frac{\psi_{j-1} - 2\psi_j + \psi_{j+1}}{a^2} = \sum_k \Delta_{jk} \psi_k,$$

where we rewrote the equation in a matrix form with

$$\Delta_{jk} = \begin{cases} -2/a^2 & \text{if } i = j, \\ 1/a^2 & \text{if } i = j+1 \text{ or } i = j-1, \\ 0 & \text{otherwise.} \end{cases}$$

We also define the discretized potential as the diagonal matrix $V_{jk} = \delta_{jk}V(ja)$.

With these the Schrödinger equation becomes

$$\sum_{k=0}^{N-1} \left(-\frac{\hbar^2}{2m} \Delta_{jk} + V_{jk} \right) \psi_k = E \psi_j.$$

This is an eigenvalue-eigenvector equation for the Hamiltonian matrix $H_{jk} = -\frac{\hbar^2}{2m} \Delta_{jk} + V_{jk}$, and can be solved efficiently using computers, most programming languages offer efficient implementations of eigensolvers.

To avoid underflow of floating-point operations it is practical to use electronvolt and nanometer units, where $\frac{\hbar^2}{2m} = 0.0380 \text{ eVnm}^2$.

1. Compute the spectrum of a particle in a box of length $L = 1 \text{ nm}$ ($V(x) = 0$) using $N = 100$ lattice points ($a = L/N$). Compare the spectrum to the analytical result. Plot the amplitude of the lowest 5 eigenstates and verify that they are approximately sine waves.
2. Compute the spectrum of a harmonic oscillator with force constant $k = 10 \text{ eV/nm}^2$ using $N = 400$ lattice points on the $-2 \text{ nm} < x < 2 \text{ nm}$ interval. You will need to shift the zero of the indexing compared to the previous case. Plot the amplitude of the ground state wavefunction, and check that its spatial extent is much smaller than the range that we simulate. This is important, because the approximation is only valid, if the wavefunction amplitude is very small at the edge of the simulated region. Compare the spectrum to the analytical result.
3. The numerical solution is useful in situations where the analytical solution is not known. Compute the ground-state energy of an electron in a quartic potential $V(x) = kx^4$ with $k = 10 \text{ eV/nm}^4$.